

AMD DEVELOPER HACKATHON: ACT II — TRACK 2

STYLEFORGE

One clip. Four voices. A Gemma that learned tone on AMD silicon.

RL fine-tuned Gemma 3 4B

Trained on AMD Instinct MI300X

Ships in a CPU-only container

Team: Sankar Subbayya x Claude (Anthropic) July 2026

The Challenge: Four Voices, One Judge

formal

News-wire precision.
Zero jokes, zero slang.

sarcastic

Dry, deadpan irony.
Never labeled, always
felt.

humorous_tech

Funny via bugs,
deploys, APIs and
rollbacks.

humorous_non_tech

Funny for everyone. No
tech allowed.

Every caption is scored by an LLM judge on two axes:

accuracy (0-1)

faithful to the clip

style match (0-1)

nails the requested tone

hidden eval set

generalize, don't overfit

10-minute wall

containerized, CPU, no
mercy

Everyone Prompts. We Train.

THE FIELD (35+ visible teams)

- Prompt a stock VLM and hope
- Tone control via instructions only
- sarcastic vs humorous blurs together — exactly what the judge penalizes

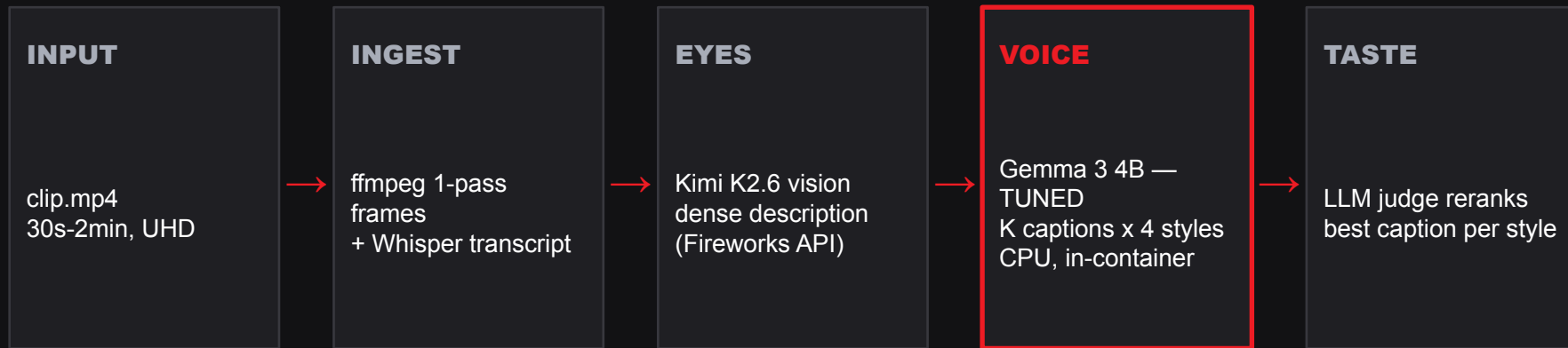
STYLEFORGE

- DPO fine-tune Gemma on judge-ranked caption pairs (RLAIF)
- Train against the same signal the leaderboard scores
- Chosen by studying ACT I's winners: the grand prize went to an RL project; fine-tuning winners paired technique with a story

Tone is trained, not prompted

DPO on judge-ranked caption pairs — optimized against the exact signal the leaderboard scores.

Architecture: Eyes, Voice, Taste



Two-stage by design: frontier eyes, trained voice. Vision never enters the training loop — the tuned model does one narrow thing perfectly: tone.

Survives judge-time: no live GPU endpoint

Every style guaranteed: 3-tier fallback ladder

No token penalty in Track 2 - BoN is free quality

The RLAIIF Data Factory

300

synthetic scenes, maximally
diverse

1,200

scene x style training cells

7,200

candidate captions by Kimi
K2.6

7,200

judge scores: accuracy +
tone

scene description → 6 candidates per style → judge scores each → best
vs worst become a DPO preference pair (quality-gap filtered)

DPO teaches Gemma: “write like the winner, not like the loser” — a thousand times over, across every style
and scene type. Judge scores are cached, resumable, and cost-logged.

Fine-Tuned on AMD Instinct MI300X

THE STACK

- 1x MI300X (192 GB HBM3) — AMD Developer Cloud
- ROCm 7.2.4 + PyTorch 2.10 (rocm/pytorch image)
- TRL: rejection-sampling SFT warm start, merged, then DPO
- LoRA r=32, bf16 — no quantization needed to train
- Merge → GGUF Q4_K_M (~2.7 GB) for CPU serving

\$1.99/hr

single-GPU droplet — credits, not cash

2.8 samples/sec

DPO throughput on a 4B LoRA

~\$4

total training cost, rehearsal included

The tuned Gemma ships inside the submission container and runs on CPU — zero GPUs, zero live endpoints, zero excuses at judge time.

Built to Survive the 10-Minute Wall

SIGKILL-PROOF OUTPUT

A valid results.json is written at startup and atomically upgraded per task. The wall can kill us mid-clip — we still score on everything finished.

DEGRADE, NEVER DIE

Best-of-N → single-shot → canned fallback, per task, per style. A dead URL, a 500ing model, malformed tasks — nothing sinks the run.

HOSTILE-INPUT HARDENED

Download deadlines beat trickling servers. Single-pass UHD frame extraction. Whisper loads once. Every clip budgeted against the clock.

TESTED LIKE WE MEAN IT

11-test suite incl. an organizer-contract simulation; multi-agent adversarial code review — 8 verified findings found and fixed.

The Eval: Measured Honestly

30 unseen scenes x 4 styles x 4 arms, scored by gpt-oss-120b (no model grades its own family)

arm	formal	sarcastic	hum_tech	hum_non_tech	ALL
Kimi K2.6 prompted	8.92	7.72	8.23	7.87	8.18
Gemma 4B base	7.88	6.78	8.18	8.02	7.72
Gemma tuned v1	7.90	7.32	7.82	7.97	7.75
Gemma tuned v2	8.40	7.13	7.93	7.83	7.83
v2 vs base	+0.52	+0.35	-0.25	-0.19	+0.11

4B matches the frontier on accuracy

After two targeted DPO rounds our 4B scores 7.62 on factual accuracy - identical to the ~1T-class Kimi (7.62). Round 1: sarcastic +0.54. Round 2: formal +0.50.

Tuning nets positive: +0.11 overall

v2 beats base 49W-38L and v1 52W-40L head-to-head. The seesaw is visible and honest: accuracy-weighted pairs traded a little humor-style sparkle for formal/sarcastic gains.

What remains is wit, not facts

The entire residual gap is the tone axis (8.03 vs 8.74): Kimi is funnier. Comedy resists distillation into 4B - so bon ships as default, tuned Gemma rides in-container.

We ship the winner — and the trained Gemma rides along: measured, not mythologized.

Gemma, Three Ways

THE VOICE — SHIPPED

DPO fine-tuned Gemma 3 4B writes every caption. Quantized to GGUF, it lives inside the container on CPU. We didn't prompt Gemma. We taught it.

THE EYES — DEMONSTRATED

Gemma 3 27B (vision) served by vLLM on the same MI300X: the full pipeline runs all-Gemma on AMD silicon, end to end, on camera.

THE PROOF — MEASURED

Training curves, before/after held-out evals, and an independent judge keeping us honest about our own model.

Deepest possible “use of Gemma” in this track: Gemma is the trained artifact — fine-tuned on AMD, shipped in the box.

STYLEFORGE

The only Track 2 entry where the model learned its voice.

Code `github.com/SankarSubbayya/styleforge`

Demo `sankarsubbayya.github.io/styleforge`

Image `ghcr.io/sankarsubbayya/styleforge:latest`

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